

Zhen Wan

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EDUCATION

Northwestern University

Master of Science in Computer Engineering

Evanston, IL

Sept 2025 – June 2027

Fudan University

Bachelor of Science in Physics

Shanghai, China

Sept 2020 – June 2025

SKILLS

Programming Languages: Python, C/C++ , Java, JavaScript, MATLAB

Frameworks: PyTorch, TensorFlow, SciPy, Matplotlib, React, Node.js, Flask

Languages: Chinese (Native), English (Fluent), Spanish (Beginner), German (Beginner)

PUBLICATIONS

UniPaint: Unified Space-time Video Inpainting via Mixture-of-Experts

Oct 2025

ICCV 1st Workshop on Human-Interactive Generation and Editing

arxiv.org/abs/2412.06340

Zhen Wan, Chenyang Qi, Zhiheng Liu, Tao Gui, Yue Ma

Backpropagation-Free Multi-modal On-Device Model Adaptation via CloudDevice

Jan 2025

Collaboration

ACM Trans. Multimedia Comput. Commun. Appl. Vol. 21, Issue 2

doi.org/10.1145/3706422

Wei Ji, Li Li, Zheqi Lv, Wenqiao Zhang, Mengze Li, *Zhen Wan*, Wenqiang Lei, Roger Zimmermann

PROJECTS

3D Reconstruction and Uncertainty Estimation of CMRI with Active Acquisition

Feb 2025 – May 2025

Undergraduate thesis, grade A. Code available at: github.com/mmmmm-w/ActiveCMR

- Led end-to-end development of *ActiveCMR*, a deployability-oriented research system that couples a conditional generative model with Bayesian uncertainty to reconstruct 3D cardiac volumes from sparse 2D segmentations.
- Designed an attention-based multi-condition aggregator to fuse heterogeneous slice conditions, improving robustness in clinical workflows.
- Collaborated with clinicians at Fudan University–affiliated hospitals to validate under real protocol settings; achieved state-of-the-art accuracy on the UK Digital Heart dataset and more than 100 retrospective clinical cases.
- Cut required acquisitions by 60 without degrading accuracy, enabling shorter exam times and higher throughput.

UniPaint: Unified Space-time Video Inpainting via Mixture-of-Experts

June 2024 – Dec 2024

Research Intern, Fudan University. Code available at: github.com/mmmmm-w/UniPaint

- Developed a diffusion model capable of performing video in-painting, out-painting, and interpolation under a single framework by generating masked areas based on given images and prior knowledge.
- Designed a Mixture of Experts (MoE) attention mechanism to enhance flexibility across video generation tasks.
- Achieved distinguished peer-review results across all tasks—equaling dedicated models, demonstrating the effectiveness of the unified model in producing high-quality output.

Multi-modal On-device Model Adaptation via Cloud-device Collaboration

June 2023 – Oct 2023

Research Intern, National University of Singapore

- Proposed a cloud-device collaboration framework that allows on-device models to adapt to their specific tasks and use cases without the need of further on-device fine-tuning.
- Utilized HyperNetwork deployed on the cloud as a fast domain adapter to generate parameters for on-device models based on device data.
- Co-authored a paper in ACM TOMM, titled “Backpropagation-Free Multi-modal On-Device Model Adaptation via Cloud-Device Collaboration”